

Referee Report

“Higher derivatives of the Gram function”

by M. Dundulis, R. Garunkštis, A. Laurinćikas and R. Šimėnas

6 August 2026

Recommendation

Major revision. The paper addresses a natural and worthwhile question — the asymptotic behaviour of the n -th derivatives of S , ϑ and the Gram function t_u — and the main asymptotic formulas appear to be *correct*: I verified Theorem 1, Corollary 2 and Theorem 3 independently for $n = 2, 3, 4$ against the classical expansions $\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi}$, $\vartheta''(t) = \frac{1}{2t}$ and $t_u \sim 2\pi u / \log u$, and the stated constants $(-1)^{n-1}(n-2)!/2\pi$, $(-1)^n(n-2)!/2$ and $(-1)^{n+1}2\pi(n-2)!$ all come out right. The Faà di Bruno bookkeeping in (17) is also correct as far as I can check.

However, the manuscript is not yet publishable. Two of the proofs contain genuine errors (one estimate in the proof of Theorem 1 is invalid as written; one step in the proof of Theorem 4 identifies a dominant term that is in fact identically zero), the hypothesis (10) underlying the universality statement is not a well-posed mathematical assumption, and Section 3 — the section on Lean and AI, which is the paper’s main selling point — contains internal contradictions and Lean code that does not type-check as printed. All of these look repairable, but they must be repaired before publication.

The detailed comments follow. Items marked **[Major]** must be addressed; items marked **[Minor]** are suggestions and corrections.

1. Errors and gaps in the proofs

1.1 [Major] The bound for $j^{(n)}(t)$ (p. 5) is invalid as written. After splitting

$$\int_0^\infty \frac{t^n(4u+1)\sigma(u) du}{((4u+1)^2 + 4t^2)^{n+2}} = \int_0^t + \int_t^\infty,$$

you bound the *first* integral by replacing $4t^2$ with $4u^2$ in the denominator. This inequality is valid (since $u < t$), but it destroys all decay in t : the resulting integral $\int_0^t (4u+1)\sigma(u)((4u+1)^2 + 4u^2)^{-n-2} du$ converges to a positive constant as $t \rightarrow \infty$, so the displayed chain only yields $j^{(n)}(t) \ll t^n$, not $j^{(n)}(t) \ll t^{-n-2}$. The estimate is therefore vacuous, and with it the error term of Theorem 1.

The fix is immediate and should be substituted: on $[0, t]$ use $(4u+1)^2 + 4t^2 \geq 4t^2$, so that

$$\int_0^t \frac{t^n(4u+1)\sigma(u) du}{((4u+1)^2 + 4t^2)^{n+2}} \ll \frac{t^n}{t^{2n+4}} \int_0^t (4u+1) du \ll t^{-n-2}.$$

The second integral, where $t^n \leq u^n$ and $(4u+1)^2 + 4t^2 \geq (4u+1)^2$ are used, is correct and gives $\ll \int_t^\infty u^{-n-3} du \ll t^{-n-2}$.

1.2 [Major] Proof of Theorem 4: the claimed dominant partition contributes the zero term. For $d^{n+1}t_u^n/du^{n+1}$ you conclude that the expression is maximised by $m'_1 = n+1$, $m'_2 = \dots = m'_{n+1} = 0$. But in Faà di Bruno’s formula with outer function $g(t) = t^n$,

that partition carries the factor $g^{(m'_1+\dots+m'_{n+1})}(t_u) = g^{(n+1)}(t_u) = 0$. The term vanishes identically and cannot be the dominant one.

Consequently the stated order is wrong. Repeating your computation with $P := \sum_j m'_j$ gives a general term of order $1/(u \log^{n+P-m'_1} u)$; the minimum of $P - m'_1$ over *non-vanishing* terms is 1, whence

$$\frac{d^{n+1}t_u^n}{du^{n+1}} = \Theta\left(\frac{1}{u \log^{n+1} u}\right), \quad \text{not } \Theta\left(\frac{1}{u \log^n u}\right).$$

The sanity check $n = 1$ settles it: the claimed formula gives $t_u'' \asymp 1/(u \log u)$, whereas Theorem 3 of this very paper gives $t_u'' \asymp 1/(u \log^2 u)$. Please also add a remark in the computation of $d^n t_u^n / du^n$ that there $P \leq n$ throughout, so no term degenerates and the conclusion $\Theta(\log^{-n} u)$ stands. The conclusion of Theorem 4 is unaffected, but the argument must be corrected.

1.3 [Major] The monotonicity argument in Theorem 4 is a non sequitur. “Based on the continuity of the $n + 1$ -st derivative, there exists some value u_0 for which $d^{n+1}t_u^n/du^{n+1}$ does not change its sign provided $u > u_0$ ” is false as stated: continuity alone implies nothing about eventual constancy of sign. What you actually need, and what your computation gives once corrected, is that $d^{n+1}t_u^n/du^{n+1}$ is asymptotic to an explicit nonzero main term of fixed sign. Please state that main term with its sign, deduce eventual monotonicity of $d^n t_u^n / du^n$, and only then invoke the criterion.

1.4 [Major] The Θ notation is used for sign-changing quantities. You define $f = \Theta(g)$ by $c_1 g(x) \leq f(x) \leq c_2 g(x)$ with $c_1 > 0$, which presupposes $f > 0$. But the quantities to which it is applied ($\vartheta^{(n)}$, $t_u^{(n)}$, the Faà di Bruno terms) alternate in sign with n . Either apply Θ to $|f|$ and track signs separately, or replace Θ by an explicit asymptotic equivalence $f \sim c g$ with the constant c (including its sign) exhibited. Since the sign is exactly what the monotonicity argument of Theorem 4 needs, the second option is preferable.

1.5 [Major] Differentiation under the integral sign is justified only for $n = 1$. You quote the Leibniz rule for $\frac{d}{dt} \int_a^\infty f$ and then assert “Evidently, they hold in our case”. For an n -fold interchange one needs, for each $k \leq n$, a t -independent integrable majorant of $|\partial^k f / \partial t^k|$ valid locally uniformly in t on $[T, \infty)$. Please exhibit such a majorant explicitly (it is easy: $|\partial_t^k [((u + 1/4)^2 + (t/2)^2)^{-2}]| \ll_k ((u + 1/4)^2 + 1)^{-2}$ for $t \geq 1$ suffices for a fixed compact t -range, together with the bound $0 \leq \sigma \leq 1/8$), and please check the reference: Rudin, *Principles of Mathematical Analysis*, 3rd ed., pp. 180–182 does not contain this statement in the form used. Cite the precise theorem (or a measure-theoretic source with dominated convergence).

1.6 [Major] Term-by-term differentiation of the series in (11) is unjustified. The expansion of $\frac{t}{4} \log(1 + \frac{1}{4t^2}) + \frac{1}{4} \arctan \frac{1}{2t}$ into $\frac{1}{16t} + \frac{1}{8t} + \sum_{n \geq 3} a_n t^{-n}$ is a Laurent series in $1/t$; state its region of convergence ($|t| > 1/2$, say) and invoke analyticity to differentiate term by term. As it stands the reader is asked to take it on faith. (The computation itself is correct: I checked $\frac{3}{16t} + O(t^{-3})$ and hence $(-1)^n \frac{3n!}{16} t^{-n-1} + O(t^{-n-3})$.) Note also that $\frac{1}{16t} + \frac{1}{8t} = \frac{3}{16t}$ should simply be combined, and it is worth remarking that the t^{-2} coefficient vanishes.

1.7 [Major] Assumption (10) is not well-posed. As printed,

$$\frac{1}{N} \sum_{k=2} |\zeta(s + it_k^n) - \zeta_{\{p \leq y\}}(s + it_k^n)|^2 < \epsilon$$

has (i) no upper summation limit (presumably $k = 2, \dots, N$), (ii) no limit operations, although the intended statement is surely of the form $\lim_{y \rightarrow \infty} \limsup_{N \rightarrow \infty} (\cdot) = 0$, (iii) an unquantified s : “with $s \in K$ ” should be $\sup_{s \in K}$ or an integral over K — the two are not equivalent and the Pańkowski argument needs one specific version, and (iv) the order of the

quantifiers “for all $\epsilon > 0$ and sufficiently large y and N ” is incompatible with the ϵ on the right-hand side. Please restate the hypothesis precisely in the exact form in which it is used in the argument of [15, pp. 192–194]. Also, define $\zeta_{\{p \leq y\}}$ before it is used, and say whether the assumption is expected to be uniform in n .

- 1.8 **[Major] The abstract promises a conjecture that the text never states as one.** The universality assertion at the end of Section 1 is an unnumbered display, derived from an unproved assumption. Please make it a numbered *Conjecture* (or “Theorem . . . under Hypothesis (10)”), and make the logical status unambiguous: what is proved unconditionally, what is conditional on (10), and what is conjectural.
- 1.9 **[Major] The criticism of Pańkowski requires substantiation.** Stating that [15] “incorrectly thought that this follows from Gallagher’s lemma” and that Sourmelidis “partially remedied this error” is a serious claim about published work and cannot stand on one sentence. Please explain precisely where the implication fails, and state exactly what Sourmelidis does and does not establish, so that a reader can see why (10) has to be assumed rather than proved.
- 1.10 **[Major] The margin in the proof of Theorem 3 is thin and should be made explicit.** Estimate (17) shows that every non-leading Faà di Bruno term is $O(u^{-(n-1)} \log^{-2} u)$, so after division by $\vartheta'(t_u) \asymp \log u$ the neglected contribution to $t_u^{(n)}$ is $O(u^{-(n-1)} \log^{-3} u)$, while the secondary term you are claiming, $2 \log \log u / \log u$ times the main term, is $\asymp \log \log u \cdot u^{-(n-1)} \log^{-3} u$. The two differ by a factor $\log \log u$ only. The theorem is therefore true, but only just, and the reader should be shown this comparison explicitly rather than left to discover it. Please also state which steps of the induction use the inductive hypothesis (13) — the leading term $m_1 = n$ uses only (8) and (9); (13) is needed only to bound the intermediate terms.
- 1.11 **[Minor] Uniformity in n .** All results are stated “for $n = 2, \dots$ ”, which invites the reading that they hold uniformly in n . They do not: the implied constants in (5), (6) and the $o(1)$ in Theorem 3 all depend on n . Please say “for each fixed $n \geq 2$, as $u \rightarrow \infty$ ”. A uniform-in- n version, if obtainable, would substantially increase the value of the paper.
- 1.12 **[Minor] Domain of definition in (7).** Solving $\vartheta(t_u) = (u-1)\pi$ with $t_u \geq 7$ requires $(u-1)\pi \geq \vartheta(7)$, i.e. $u \geq \vartheta(7)/\pi + 1$. The stated bound $u \geq \vartheta(7)/\pi + \pi$ appears to be a typo (and is dimensionally odd). Please also state the relation to the classical Gram points: with $\vartheta(g_m) = m\pi$ one has $t_k = g_{k-1}$; readers will otherwise be confused by the index shift.
- 1.13 **[Minor] Where do the derivatives of S exist?** Since (2) holds on each interval (γ, γ') between consecutive ordinates, Theorem 1 should say explicitly that $S^{(n)}$ is taken on the complement of the (countable, closed-discrete) jump set, and that the O -constant is uniform over that complement. In Corollary 2 an extra sentence is needed: the identity $\vartheta^{(n)} = -\pi S^{(n)}$ holds a priori only off the jump set, and one extends it to all $t \geq T$ by continuity of $\vartheta^{(n)}$.
- 1.14 **[Minor] Corollary 5 does not follow from Theorem 4.** On p. 3 you write that it “follows directly from Theorem 4”, but the proof on p. 8 does not use Theorem 4 — it re-applies the derivative criterion to $f(k+t)$ and $f(kt)$. Please reconcile. Moreover the verification of the two hypotheses of [11, Ch. 1, Thm. 9.6] is asserted rather than performed; for $g(x) = f(xt)$ one has $g^{(n)}(x) = t^n f^{(n)}(xt)$ and $x|g^{(n)}(x)| = t^{n-1} \cdot (xt)|f^{(n)}(xt)| \rightarrow \infty$ for each fixed $t > 0$ — half a line, but it should be there, along with the remark that $t = 0$ is a null set.
- 1.15 **[Minor] Novelty should be delineated.** Theorem 4 with $n = 1$ (uniform distribution mod 1 of the Gram points) is classical; Corollary 5 is, as you say, already in [12, Lemma 5.7]. Please add a short paragraph stating precisely what is new: as I read it, this is Theorem 1 and Corollary 2 for $n \geq 2$, Theorem 3 for $n \geq 3$ together with the corrected constant for $n = 2$, and Theorem 4 for $n \geq 2$.
- 1.16 **[Minor] The correction to [10, Lemma 1.1] is asserted without evidence.** “Here,

we corrected the constant in [10, Lemma 1.1] for $n = 2$ is a claim that a reader cannot check. Quote Korolev’s statement, display your own, and say in one sentence where the discrepancy originates.

2. Section 3 (Lean and AI)

This section is what distinguishes the paper from a routine calculation, and it is currently the weakest part. In its present form a reader cannot tell what has actually been verified.

- 2.1 [Major] Two mutually contradictory claims about axioms.** “Our axiomatization is minimal, consisting of the results in the theory of zeta functions and uniform distribution” (p. 4) versus “The project builds cleanly, with no additional axioms apart from the standard mathematical ones and no `sorry` statements” (p. 8). These cannot both be true. If the analytic input (monotonicity of ϑ on $[7, \infty)$, the Riemann–von Mangoldt formula, the representation (3) of δ , the properties of ρ and σ) is axiomatised, then the Lean development verifies the *derivations*, not the theorems, and this must be said plainly. Please include a complete, explicit list of the axioms, ideally as a printed table with the corresponding classical references, and report the output of `#print axioms` for each of the five main results.
- 2.2 [Major] The displayed Lean code does not type-check as printed.** `S` is declared as `S (F : StepFunction) (t : Real) : Real`, i.e. it takes a step function as its first argument, but `theorem1` uses the bare identifier `S` inside `iteratedDeriv n S t`. Either the definition or the theorem statement is mis-transcribed. Likewise `theta`, `gramLeading`, `gramPow`, `phi` and `delta` are used but never defined in the text; `gramLeading` in particular carries the entire content of Theorem 3 and must be displayed.
- 2.3 [Major] Theorem 3’s formalisation is strictly weaker than Theorem 3.** The informal statement $t_u^{(n)} = \frac{(-1)^{n+1} 2\pi(n-2)!}{u^{n-1} \log^2 u} (1 + (2 + o(1)) \frac{\log \log u}{\log u})$ asserts that the remainder after subtracting the main term and the $2 \log \log u / \log u$ correction is *little-o* of `gramLeading` · $\log \log u / \log u$. The Lean statement uses `Is0` (big- O), which is a genuinely weaker assertion and does not pin down the constant 2 — precisely the constant you claim to be correcting in [10]. Please use `IsLittle0`, or explain why big- O is the intended formalisation and adjust Theorem 3 accordingly.
- 2.4 [Major] Function.invFunOn produces junk without injectivity.** `gram` is defined as `Function.invFunOn theta (Set.Ici 7) ((u-1)*Real.pi)`, which is meaningful only if ϑ is proved injective on $[7, \infty)$ and if the value $(u-1)\pi$ is proved to lie in $\vartheta([7, \infty))$. Otherwise `gram` returns an unspecified element and `theorem3` may be vacuous or accidentally provable. Please state whether ϑ ’s monotonicity on $[7, \infty)$ is proved or axiomatised, and whether the defining property $\vartheta(\text{gram } u) = (u-1)\pi$ is available as a lemma on the relevant domain. Note also that `theorem3` imposes no lower bound on u , whereas the informal statement needs $u \geq \vartheta(7)/\pi + 1$; the filter `N^{\infty}_0` presumably handles this, but it should be explained.
- 2.5 [Minor] Notation and typography of the listings.** `Iso` in `theorem3` is a typo for `Is0`. The listings contain escaped L^AT_EX (`\forall`, `\in`, `=^f[nhds s]`) rather than the Unicode Lean 4 syntax that the compiler accepts; please reproduce the source verbatim (a `listings/minted` environment with a Lean lexer will do). Mathlib’s identifier is `Asymptotics.IsBig0`, usually written with the notation `=0[atTop]`; if `Is0` is your own abbreviation, say so. The filter `N^{\infty}_0` is never defined, and the gloss “ $f(x) = O(g(x))$, $n \rightarrow \infty$ ” mixes up the variables ($t \rightarrow \infty$ is meant).
- 2.6 [Minor] Reproducibility.** A mutable GitHub URL is not a citable artefact. Please give a commit hash and archive the repository (Zenodo DOI or similar), state the Mathlib revision as well as the Lean version (v4.29.1 alone is not enough), and give build instructions and an approximate build time. “More than 10000 lines” is not a measure of quality; a table of

files, main declarations, and axioms used would be far more informative. The URL is given twice (Sections 1 and 3); once suffices.

- 2.7 [Minor] Say what is *not* formalised.** Is the classical criterion from Kuipers–Niederreiter Ch. 1 Thm. 3.5 proved, or assumed? Are Fejér’s theorem and the van der Corput bound proved from scratch (the text suggests yes — this is a real contribution and deserves to be highlighted)? Is the conditional universality statement formalised at all? A short table “result / formalised / axiomatised / not treated” would answer all of this at a glance.
- 2.8 [Major] Report whether formalisation caught any errors.** This is, to my mind, the most interesting scientific question the paper could answer, and it is currently unaddressed. Two of the informal proofs contain errors (items 1.1 and 1.2 above). Did the Lean development reproduce those steps, and if so, how were they discharged? If the Lean proof of Theorem 1 bounds $j^{(n)}$ correctly while the paper does not, say so — it is a concrete demonstration of the value of formalisation. If the Lean development avoids the issue by axiomatising around it, that is important for the reader to know.
- 2.9 [Minor] AI-usage disclosure.** The abstract’s “We use AI to prove our results in Lean” overstates matters and should be softened (the AI assisted in producing formalisations; the proofs are yours and the responsibility is yours). Please add a short methodology paragraph: which model was used for what, how output was checked, the rough proportion of machine-generated versus hand-written code, and a statement that the authors have verified the final artefact. Model designations should be dated, since they are not stable identifiers. Most journals now have a formal AI-disclosure policy; please check the target journal’s.
- 2.10 [Minor] “Lean has emerged as the definitive programming language for proving mathematical theorems”** is an overstatement given Rocq/Coq, Isabelle/HOL and Mizar. “One of the most widely used” or “the system with the most extensive contemporary mathematical library” would be defensible.

3. Presentation, notation and references

- 3.1** Notation clash: n denotes the order of differentiation throughout, but is reused as the summation index in $\sum_{n \geq 3} a_n t^{-n}$ in the proof of Theorem 1. Use a different letter. Similarly $N(t)$ (zero-counting function) collides with N (the length of the sum) in (10) and in the universality statement.
- 3.2** The display $\left(-\frac{t}{2\pi} \log \frac{t}{2\pi} + \frac{t}{2\pi} - \frac{7}{8}\right)^{(n)} = \left(-\frac{1}{2\pi t}\right)^{(n-2)}$ abuses notation badly — the right-hand side reads as a power. Write $\frac{d^{n-2}}{dt^{n-2}}\left(-\frac{1}{2\pi t}\right)$, and note the step needs $n \geq 2$.
- 3.3** “Faà di Bruno” and “Faà de Bruno” both appear; standardise. Please also double-check the Gradshteyn–Ryzhik formula numbers 0.42 and (0.430).
- 3.4** Redundancy on pp. 1–2: the Riemann–von Mangoldt formula is introduced twice in almost identical words (“The Riemann-von Mangoldt formula ... leads to the following relation” and “The function S is related to the function N by the Riemann-von Mangoldt formula”). Merge.
- 3.5** Simplify $\frac{2\pi}{2u \log u}$ to $\frac{\pi}{u \log u}$ in the proof of Theorem 3; similar unreduced fractions occur elsewhere.
- 3.6** In (16) the two extreme terms are separated by an ellipsis that hides the structure. Display the full Faà di Bruno sum once, then name the two terms.
- 3.7** “Asymptotically largest” is used repeatedly without definition; given the Θ -issue in 1.4, define it once, in terms of absolute values.
- 3.8** Add MSC 2020 classification numbers and keywords; most journals in this area require them.
- 3.9** Consider adding a short table of numerical values of $t_u^{(n)}$ for $n = 2, 3, 4$ against the asymptotic

formula. For results of this kind it is cheap, convincing, and would have caught the sign/order slip in item 1.2 immediately.

3.10 Possible strengthening (optional, but it would raise the paper’s interest considerably): the shape $1 + (2 + o(1)) \frac{\log \log u}{\log u}$ suggests a full asymptotic expansion in powers of $\frac{\log \log u}{\log u}$, with an explicit error term. Since the machinery is already in place, deriving the next term would be a natural addition.

3.11 References:

- Edwards “pp. 125–226” is presumably “125–126”.
- Ref. [15] (Pańkowski): stray “1” in “45(1):181–195, 1 2018”.
- Ref. [11] (Kuipers–Niederreiter): give the original Wiley 1974 edition or the Dover reprint properly; “Courier Corporation, 2012” is unhelpful, and the theorem numbers 3.5 and 9.6 should be checked against the edition cited.
- Add DOIs throughout; verify the volume/issue data of Ref. [10].
- Gram’s law: the statement that Hutchinson “found counterexamples” is correct but thin; a sentence recording that Gram’s law fails infinitely often (Titchmarsh), with the relevant reference, would place the paper better.

3.12 Minor language edits are needed throughout (e.g. “The first member in the above equation” → “the first term”; “the last one is asymptotically the largest of the remaining ones” needs a comma-free rewrite). A careful copy-edit pass is advisable.

4. Summary

The mathematical results are, as far as I can verify, correct and worth publishing; the constants check out on independent computation for small n , and the Faà di Bruno bookkeeping is sound. The Lean/AI component is a genuinely interesting addition. But the paper cannot appear as it stands: the estimate for $j^{(n)}$ must be redone (1.1), the dominant-partition argument in Theorem 4 must be corrected (1.2, 1.3), hypothesis (10) must be given a precise formulation (1.7), the criticism of [15] must be substantiated (1.9), and Section 3 must be rewritten so that the reader can determine what has actually been proved in Lean and what has been assumed (2.1–2.4, 2.7, 2.8).

I would be glad to look at a revised version.